

Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue
Alexandria, VA 22314

Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan

The Oxford Martin AI Governance Initiative (AIGi) welcomes the opportunity to submit this comment for consideration in the development of the United States' AI Action Plan, as called for in Executive Order 14179. AIGi supports the Administration's vision of American leadership in artificial intelligence as a driver of economic prosperity and national security. We propose a policy framework to solidify the U.S. position as the global leader in AI while addressing a restricted set of security risks to American citizens and interests.

This comment focuses on three key areas where targeted government action can have the greatest impact:

1. **Strengthening US AI Leadership and Competitiveness** through strategic investments in critical infrastructure, compute capacity, and research priorities
2. **Protecting US Citizens and National Security** by eschewing premature, innovation-dampening regulation in favor of targeted, capability-based security governance and robust measures against AI system vulnerabilities
3. **Managing Access to Strategic AI Capabilities** through effective export controls and thoughtful approaches to open-source AI development

About the AI Governance Initiative: The Oxford Martin AI Governance Initiative (AIGi) is an interdisciplinary research center dedicated to understanding and anticipating lasting risks from AI through rigorous research. Housed at the University of Oxford's Martin School, we combine technical expertise with policy analysis to support decision-makers across industry, government, and civil society in mitigating AI's challenges while realizing its benefits.

As a UK institution, we strongly support American leadership in AI development, recognizing our shared values and commitment to ensuring advanced AI systems are built and governed by nations that uphold free speech, human rights, and transparent governance. The transatlantic partnership on AI governance is essential to ensure these technologies serve humanity's best interests.

By Charles Martinet, Ben Harack, and Sumaya Nur Adan, research affiliates at the AI Governance Initiative. The views expressed in this comment do not necessarily reflect the positions of individual authors. Correspondence with

Introduction

The development of artificial intelligence has entered a phase of extraordinary acceleration and global diffusion, creating unprecedented opportunities and challenges for American economic prosperity and national security. The U.S. AI Action Plan represents a critical opportunity to shape the trajectory of this transformative technology in ways that serve American interests and values.

Recent developments create three critical imperatives for U.S. policy. Infrastructure bottlenecks—from energy supply to specialized semiconductor availability—threaten to constrain America's development capacity precisely when computational resources have become a strategic asset. Novel security vulnerabilities unique to AI systems require targeted governance measures that scale with system capabilities without hampering innovation. And the global proliferation of AI technologies necessitates managing global access to the most sensitive capabilities.

Without strategic government action addressing these interconnected challenges, the United States risks losing both its technological edge and its ability to shape how these powerful systems evolve. The recommendations that follow provide a framework for securing U.S. economic and security leadership in AI.

1. Strengthening US AI Leadership and Competitiveness

a. Critical Energy Infrastructure for AI Leadership: Streamline approvals and incentivize development of abundant, reliable energy infrastructure to power America's AI revolution.

Access to reliable power is becoming strategically important in AI leadership. The demand for computing power required for advanced AI systems is projected to increase dramatically as demand for increasingly capable models grows. Matching this demand will require much more power than current infrastructure can provide—and projected growth in energy production is not expected to match the growth in demand.

Ensuring abundant, cost-effective electric power supply will allow American AI developers to continue to hold a crucial competitive advantage over their international rivals. Targeted tax incentives and streamlined approvals for energy providers could accelerate private investment in necessary infrastructure, creating a foundation for robust AI development at scale and preventing energy shortages from becoming a bottleneck that constrains America's technological leadership.

Recommendations:

- Create expedited approval pathways and tax incentives for energy projects supporting AI-specialized data centers.
- Establish a federal task force to coordinate energy planning for high-priority AI development zones

b. Data Center Development: Ensure that America remains the leading location for AI compute through strategic incentives, expedited permitting, and infrastructure investments.

Domestic data center capacity represents one of the most significant potential bottlenecks in advancing American AI leadership. The federal government should ensure that specialized facilities required for large-scale AI training do not face complex regulatory hurdles and lengthy approval processes that impede rapid expansion. Without deliberate policy support, data center construction timelines and costs may place American AI developers at a disadvantage compared to international competitors who benefit from streamlined approval processes and government support.

Expedited permitting could dramatically accelerate deployment of necessary computing resources, ensuring that the United States maintains its position at the forefront of AI compute capacity. Prioritizing these facilities in regions with abundant renewable energy would create AI development hubs that maximize both economic and environmental benefits while strengthening America's technological leadership.

Recommendations:

- Streamline review processes and create tax incentives for high-capacity AI data centers serving national security or strategic economic purposes (for example through "AI Compute Zones").
- Establish matching grants for data center developments in regions with abundant renewable energy resources.

c. Strategic Semiconductor Supply Chains: Secure preferential access to advanced AI chips through deepened partnerships with allies while strengthening targeted domestic capabilities.

Advanced AI-specific semiconductors represent a critical vulnerability in America's AI supply chain, with global production concentrated in few, often geopolitically sensitive regions. The technical complexity and capital-intensive nature of leading-edge semiconductor manufacturing make complete domestic production infeasible in the near term. The U.S. should grow onshore semiconductor capacity through tax incentives, streamline regulation, and public-private partnerships. In addition, because increasing domestic capacity is a long-term project, the United States should leverage America's design expertise while securing reliable access to manufacturing capacity through strategic international partnerships.

The US should pursue deals to strengthen AI supply chain resilience, in particular agreements to secure preferential access to critical components. By receiving the majority of global AI-specialized chips, the US can maintain its leadership position in AI development. This strategy complements domestic data center development, which should remain a priority regardless of semiconductor sourcing decisions, creating a comprehensive approach to compute infrastructure security.

Recommendations:

- Implement policies ensuring American companies receive preferential access to advanced AI chips produced globally.
- Deepen partnerships with trusted semiconductor manufacturing nations to secure supply chain stability. Work with Japan, South Korea, Taiwan, and European partners to 1) coordinate R&D of next-generation semiconductor designs and materials and 2) pursue a "solidarity pledge" ensuring no party will cut off others from critical equipment, design IP, or scarce AI-specialized chips.
- Enhance capacity for packaging, testing, and assembling advanced semiconductors within the United States.

d. Strategic AI Research & Development Initiatives: Fund targeted R&D programs that enhance U.S. security interests while developing targeted governance tools and technical standards.

Many current AI governance approaches lack robust technical foundations; without rigorous evaluation methods, standardized testing protocols, and technically-validated security measures, they may lead to regulatory regimes that create security vulnerabilities and inhibit innovation. The U.S. government should strategically invest in R&D initiatives that simultaneously strengthen America's competitive advantage and enable evidence-based policy with respect to targeted security risks. By supporting fundamental research in areas such as model evaluation, security mechanisms, and next-generation hardware, the government can address market failures where private industry lacks sufficient incentives to invest. Such efforts can also address areas where industry players are prohibited by antitrust rules from collaborating on valuable industry-wide initiatives. These investments will position the U.S. to shape global technical standards and help identify and mitigate emerging security vulnerabilities before they can be exploited by adversaries.

Recommendations:

- Allocate funding for the research and development of:
 - 1) rigorous, standardized model security evaluation techniques;
 - 2) hardware security mechanisms to verify computational resource usage, facilitate privacy-preserving access to datasets, and enable reliable attestation of evaluation results, helping ensure that high-end AI hardware is used in accordance with national security interests;
 - 3) adversarial testing and resilience measures against manipulation, theft, and misuse of advanced AI systems;
 - 4) technical guardrails for AI systems that reduce security risks without compromising America's leadership in innovation; and
 - 5) next-generation AI hardware research to ensure US leadership in AI infrastructure.
- See [Reuel, Bucknall, et al., 2024](#) for key R&D priorities.
- Ensure these R&D initiatives directly inform the development of practical governance tools and standards that can be operationalized by industry and government alike.

- Create shared research infrastructure enabling secure, controlled access to models and datasets for legitimate security research while protecting intellectual property.
- Expand partnerships with key allies to jointly fund and develop AI security research, ensuring safe technological advancements in AI governance and security evaluation.

e. A US Lead in Secure and Verifiable AI: Provide market-creating standards that help American businesses lead in providing security-sensitive AI services to people, companies, and states.

Absent regulatory effort, uptake of AI will lag in security-sensitive domains such as medicine, government services, and militaries. Organizations in these sectors require verifiable assurances that AI systems meet stringent security and reliability standards before deployment. Technological stacks already exist that can robustly verify that AI models and their infrastructure follow desired rules—but the trustworthiness of these assurance systems is limited by the lack of a neutral overseer who can validate security claims and ensure compliance with technical requirements.

Federal regulation can enable this market in at least two major ways. First, it can provide an institutional framework for strong, independent, and neutral oversight of secure and verifiable AI stacks, thus enabling American corporations to robustly demonstrate their cutting-edge offerings in this domain. Second, it can minimize compliance costs by facilitating the creation of targeted industry standards for secure and verifiable AI. Compliance costs can be reduced by ensuring that all federal rules about technical aspects of AI systems are compatible with and do not overreach the new standard. Broad use of this standard would help mitigate the potential costs for AI producers from regulatory fragmentation and existing regulatory overreach.

Rapid progress on the standard and its rollout can also enable US-based AI infrastructure providers to lead in this emerging domain on the international level. Many states will desire domestic data centers that meet security standards strict enough that they can be allowed to locally host sensitive resources such as models exported from the USA. By establishing the technical standards and certification processes for these secure facilities, American companies can capture a significant share of this growing market while advancing U.S. security interests. American leadership in this domain will provide another competitive edge in the emerging AI ecosystem, positioning U.S. companies as the premier providers of secure AI solutions globally.

Recommendations:

- Direct NIST to develop a Secure and Verifiable AI Framework (SVAF) in close partnership with industry leaders, establishing how an organization can make credible technical claims about its infrastructure and software such that they can be securely verified.
- Create a neutral oversight body that can verify claims in accordance with the SVAF, thus allowing AI producers to robustly demonstrate the security and verifiability of their offerings.

- Extend SVAF rules to enable foreign data centers and AI industry players who want to demonstrate that they meet American standards for security and verifiability as part of their eligibility for U.S. government contracts or hosting models exported by the U.S.
- Collaborate with key allies to ensure international recognition of US verification standards, securing reciprocal agreements for market access based on comparable security assurances.

2. Protecting US Citizens and National Security

a. An AI Security Governance Framework Based on Capability Tiers: Develop and implement targeted governance requirements that scale with AI capabilities rather than applying blanket regulations that could hamper America's competitive edge.

The accelerating pace of AI development demands a governance approach that addresses genuine risks while preserving the innovation ecosystem that fuels American leadership. Applying uniform regulations across all AI systems regardless of their capabilities and risk profiles would hamper innovation and competitiveness. Instead, a framework based on evidence-based capability tiers would match governance requirements to the specific technological capabilities and risk profiles of AI systems, focusing on those with the greatest security impacts.

Without capability-based distinctions, regulators risk either imposing excessive burdens on low-risk systems or failing to adequately oversee high-capability applications that could affect national security—potentially driving innovation offshore as developers seek jurisdictions with less blunt regulatory approaches. Instead, requirements should apply only to systems that demonstrate specific security-related capabilities of concern, creating regulatory predictability for developers and allowing them to anticipate compliance requirements as their systems advance.

Leading AI developers have already implemented voluntary security and safety measures that can form the foundation for a more formal capability-based framework. These include conducting internal and external red-teaming to identify severe threats, establishing information sharing protocols to alert others to emerging vulnerabilities, implementing cybersecurity and insider threat safeguards to protect model weights, enabling third-party vulnerability reporting, and other best practices that help reduce security risks and enable uptake of AI solutions.

The federal government should foster industry consensus on these practices, for example through the US AI Safety Institute Consortium, to establish baseline expectations appropriate for systems of different capability levels. It should also lead efforts to internationalize these security practices through both diplomatic and regulatory channels to prevent regulatory arbitrage while also expanding market opportunities for American AI products. By establishing clear, capability-based requirements that are adopted by key international partners, the U.S. can create a coherent framework that enables responsible innovation while addressing security concerns.

Recommendations:

- Build government capacity to independently monitor and evaluate AI capabilities, trends, and emerging risks through appropriate agencies and departments (see our recommendation for the AI Safety Institute), with the view of leveraging this capability to develop a framework based on capability tiers.
- Task an inter-agency taskforce with developing a tiered governance structure based on clear technological thresholds or capability tiers that, when reached, would trigger specific policy responses, focusing oversight on systems with the greatest potential for misuse and loss of control risks. The taskforce should define specific milestones and the broad contours of the policies to be implemented after each milestone.
- Require companies developing advanced AI systems to publish or report to the government appropriate documentation of their capability assessment frameworks and risk mitigation strategies, helping the federal government develop appropriate targeted requirements.
- Internationalize capability-based security standards to ensure a global level playing field by leading development of technical specifications through bodies like ISO and NIST, embedding these standards in trade agreements with key partners, and creating certification mechanisms that verify compliance—potentially limiting market access for non-compliant AI systems in critical sectors.

b. Public-Private Partnerships & Information Sharing: Establish robust communication channels between government and industry to address evolving AI security challenges.

The rapid evolution of AI capabilities and potential threats requires unprecedented coordination between government and industry. The technical complexity of these systems and the specialized expertise required to evaluate them necessitates close collaboration between those developing the technology and those responsible for securing national interests. Government agencies lack the technical depth to independently evaluate cutting-edge AI systems without industry cooperation, while private developers may be unaware of emerging threat vectors identified through intelligence channels. Without formalized information sharing, both sectors will face critical knowledge gaps and potential security blind spots that could be exploited by adversaries seeking to compromise these systems.

These communication channels would help build institutional capacity within government while enabling a coordinated response to emerging threats, strengthening the United States' ability to identify and mitigate AI security risks before they materialize. Establishing trusted channels for sharing sensitive information about vulnerabilities, incidents, and mitigation strategies would create a more resilient AI ecosystem capable of withstanding sophisticated attacks and addressing security challenges as they emerge.

Recommendations:

- Establish formal communication channels with AI developers, similar to Information Sharing and Analysis Centers (ISACs) in other critical industries.
- Create both classified and unclassified channels to share threat information and security practices and include intelligence agencies as and when appropriate in information-sharing frameworks.
- Develop secure government infrastructure for assessing sensitive AI technologies that may have national security implications, enabling thorough assessment of potentially sensitive AI technologies without exposing evaluation methodologies or sensitive capabilities to potential adversaries

c. Crash Program for AI Model Security: Launch a coordinated national effort to secure the most capable AI systems across their entire lifecycle—from training environments to deployed models—against attack, theft, and manipulation.

As AI systems become increasingly integrated into critical infrastructure and sensitive applications, they present novel and potentially severe security vulnerabilities. The unique characteristics of these systems create security challenges that existing cybersecurity frameworks are ill-equipped to address. Inadequate security of AI assets—from physical data centers to model weights—represents one of the few vulnerabilities that could rapidly erode America's current AI leadership position. Existing security approaches will prove insufficient for protecting these systems from sophisticated attacks designed to compromise their integrity or extract sensitive intellectual property.

Security vulnerabilities exist across multiple levels, including physical infrastructure (data centers, power systems), hardware (specialized chips), training data, model weights, algorithms, and deployment environments. These systems face diverse threat vectors including cyber intrusions targeting model weights or training data, physical security breaches at AI infrastructure facilities, insider threats from personnel, compromises of AI hardware components, and advanced adversarial attacks unique to machine learning systems. This multifaceted attack surface requires a comprehensive security approach that addresses vulnerabilities at each level.

Market forces alone cannot adequately address these security challenges due to information asymmetries between developers and users (where developers have greater knowledge about vulnerabilities but limited incentives to disclose them) and antitrust constraints limiting necessary industry-wide coordination. Individual companies lack sufficient incentives to invest in security measures that provide collective benefits, particularly when competitive pressures encourage rapid deployment over thorough security validation. Without coordinated federal action to establish security standards and practices, the commercial AI ecosystem cannot achieve the necessary security posture to withstand determined adversaries.

A national initiative should involve collaboration between national security agencies, research institutions, and private sector AI developers to rapidly identify and address emerging vulnerabilities. This approach would leverage the federal government's security expertise and research capabilities as well as industry's

technical knowledge to create robust security standards and practices appropriate for systems of different capability levels. Establishing American leadership in AI security would create both national security advantages and commercial opportunities in secure AI applications, positioning U.S. companies as trusted providers in domains where security is paramount.

Recommendations:

- Fund targeted research into novel vulnerability types unique to the most capable models, particularly those that could enable model weight theft or compromise.
- Develop testing methodologies to evaluate model security against adversarial attacks and unauthorized access across different capability tiers, ensuring that cyber and physical security measures scale appropriately with AI system capabilities (see our recommendation on a capability-based security governance framework).
- Establish joint efforts between government agencies and leading AI companies to rapidly identify baseline security standards appropriate for systems of different capability levels and address emerging security vulnerabilities at all levels.
- Require security assessments proportional to capability levels before deployment and, in some cases, development of the most capable systems.

d. The AI Security Institute: Transform the existing AI Safety Institute into a center of technical excellence that enhances government capacity to evaluate, secure, and verify advanced AI systems while coordinating international security standards with trusted partners. The institute should be renamed the AI Security Institute (AISI), and focus exclusively on national security concerns. This approach will safeguard critical resources, keeping them focused exclusively on essential security imperatives.

The federal government currently lacks sufficient technical capacity to evaluate emerging AI systems, assess their security implications, and develop appropriate policy responses. This capability gap undermines the government's ability to make informed decisions. Without robust internal expertise, the government will be unable to effectively respond to potential threats as they emerge, potentially leaving critical vulnerabilities unaddressed until after they are exploited.

The institute should develop rigorous testing methodologies, evaluate both domestic and foreign AI models in terms of capabilities, risks, and vulnerabilities, and provide technical guidance to government agencies. It must function as a critical node within a broader network of federal offices to maximize economic and national security benefits, coordinating with defense, intelligence, and civilian agencies to ensure a coherent approach to AI security. These activities would create a foundation of technical expertise within government capable of understanding emerging capabilities and their security implications.

Building this capacity will provide the U.S. government with crucial situational awareness regarding AI capabilities being developed domestically and abroad while strengthening American influence over the global

direction of AI governance. Without American leadership in international standards development, frameworks developed by other nations may disadvantage U.S. companies or prove inadequate for addressing security concerns that impact American interests. To ensure such standards reflect American security priorities, the Institute needs to develop credible technical expertise and evaluation methodologies.

Recommendations:

- Reframe the AISI's mission to focus on building government capacity to evaluate AI systems, including those developed by potential adversaries.
- Secure appropriate funding from Congress to attract top technical talent and build the infrastructure necessary for advanced testing and verification processes.
- Develop comprehensive model weight security protocols and verification methodologies to protect critical AI assets from theft or exploitation.
- Coordinate testing and evaluation methodologies with similar institutions in allied nations to establish consistent and scalable international approaches to AI security that will minimize compliance costs across markets.
- Share appropriate insights on emerging capabilities and risks with trusted partners to strengthen the collective security posture of allied nations.
- Coordinate technical assessment efforts across relevant federal agencies to optimize threat response capabilities and establish clear coordination mechanisms with national security authorities, including dedicated liaison positions.

e. Minimum and Maximum Federal Standards: Establish clear federal parameters for AI regulation that provide consistency for developers and mitigate the danger of a harmful patchwork of conflicting state requirements.

The emergence of inconsistent state-level AI regulations could create compliance challenges for developers and potentially hamper innovation. As states develop their own approaches to AI governance, developers may face contradictory requirements across different jurisdictions, increasing compliance costs and creating regulatory uncertainty that discourages investment. Without federal leadership to establish coherent national standards, this regulatory fragmentation could place American developers at a disadvantage compared to international competitors operating under more coherent frameworks.

A strategically designed federal framework would provide the regulatory certainty needed for long-term investment while ensuring that essential safety guardrails remain consistent nationwide. By establishing clear parameters for state regulation, federal policy can prevent harmful inconsistencies while preserving appropriate flexibility for state innovation. Such a framework should recognize that some domains benefit from regulatory uniformity (maximum harmonization preventing states from imposing stricter requirements), while others warrant flexibility for states (minimum harmonization establishing federal floors rather than ceilings).

Federal policymakers can use maximum harmonization to shield lower-risk AI innovations from a patchwork of conflicting state regulations, while using minimum harmonization to ensure baseline protections for higher-risk systems. Furthermore, the very structure of state regulation might be subjected to minimum harmonization, where federal rules could ensure that state regulation integrates cleanly with the established federal rule structures for AI, such as the SVAF described above—thus ensuring that compliance processes are scalable and cost-effective for industry even if many states implement their own requirements. This approach recognizes legitimate state interests and the vital role they play as policy laboratories where regulatory innovations can be tested and refined, while preventing harmful fragmentation that would place American companies at a disadvantage against international competitors operating under more unified regulatory frameworks.

Recommendations:

- Work with Congress to enact legislation that clearly designates areas where states are preempted from imposing additional regulations (maximum harmonization) and areas where federal standards serve as a floor that states may build upon (minimum harmonization).
- Prioritize maximum harmonization for lower-risk AI applications to create a unified national market for responsible innovation, while employing minimum harmonization for applications with potential for significant harm.
- Particular attention should also be paid to the scalability and interoperability of rules through standards such as the SVAF so that industry can cost-effectively meet its obligations to both states and the federal government.

3. Managing Access to Strategic AI Capabilities

a. Strategic Computing Access Framework: Strengthen America's approach to controlling advanced AI technology transfer while maintaining global leadership through selective engagement.

The existing Framework for AI Diffusion (BIS-2025-0001) established important groundwork for managing access to sensitive AI technologies, but requires substantial enhancement to address emerging vulnerabilities across both hardware and cloud domains.

Unlike semiconductor chips that cannot be recalled once shipped, cloud access can be monitored in real-time, throttled depending on usage patterns, and revoked entirely if necessary. This dynamic nature allows the United States to implement more precise restrictions while maintaining the ability to quickly respond to changing security concerns. Cloud controls can be calibrated based on specific use cases, specific capabilities being accessed, and evolving risk assessments, creating a more adaptive security framework than approaches focused on hardware exports alone.

A robust, multi-layered system of controls is essential to prevent adversaries from accessing sensitive AI capabilities while enabling appropriate collaboration with trusted partners and allies. Strategic governance must encompass both hardware exports and cloud compute access, recognizing these as interconnected vectors through which critical capabilities can flow. Without cohesive coordination between these complementary control mechanisms, sophisticated actors will inevitably exploit regulatory gaps to circumvent restrictions, rendering individual control mechanisms less effective, and redundant regulations risk imposing unnecessary economic costs.

Maintaining strong international cooperation on hardware export controls must remain a top priority, as the effectiveness of US-led restrictions depends on alignment with key allies to prevent circumvention through alternative supply chains. While significant progress has been made in coordinating hardware export controls with Japan and the Netherlands, equivalent frameworks for cloud-based AI compute with key countries remains underdeveloped. Domestic data centers pose less concern due to direct US regulatory oversight, and trusted Tier 1 allies generally maintain compatible security standards; but restricted entities can still bypass restrictions by accessing equivalent capabilities through foreign cloud providers operating under less stringent regulatory frameworks. Harmonizing these approaches is essential to creating a coherent international control regime that prevents capability leakage while maintaining America's technological leadership. In this regard, strategic engagement with key developing economies represents both an opportunity to maintain American technological reach and a necessity to counter China's growing digital influence in these regions. By establishing structured technology partnerships with select developing nations, the U.S. can expand its commercial footprint while ensuring these countries are not driven toward alternative technology ecosystems that could undermine American interests.

Recommendations:

- Further differentiate access tiers in the Framework for AI Diffusion based on comprehensive security and economic assessments:
 - Deepen technology sharing with closest allies (Tier 1) who maintain equivalent control systems
 - Establish conditional access for other partners (Tier 2) based on verifiable commitments to 1) responsible use, 2) enforcement of anti-chip smuggling rules and U.S. export control laws, and 3) implementation of robust datacenter governance standards that align with future US-led efforts to establish interoperable international frameworks for cloud compute oversight
 - Strengthen capacity to monitor the effectiveness of restrictions for countries of concern (Tier 3)
 - Develop clear criteria and processes for countries to navigate between tiers through specific policy reforms and security commitments, with the understanding that Tier 2 may be further subdivided as the control framework matures
- Strengthen end-user verification protocols to track both physical hardware transfers and cloud resource usage to eliminate regulatory blind spots.

- Work with key technology partners to develop a comprehensive framework that addresses both cloud computing access and semiconductor export controls as complementary but distinct mechanisms, implementing consistent thresholds and monitoring protocols to close regulatory gaps and ensure that American cloud providers are not disadvantaged against allied competitors operating under lighter regulatory standards.

b. Strategic Framework for Securing Open Source AI: Develop a nuanced, capability-based approach to open-source AI that balances innovation, security, and America's global leadership position.

Once the weights of an AI model are made widely available, this action is irreversible—they cannot be recalled, modified, or meaningfully controlled in their spread. This irreversibility represents a unique security challenge. Unlike traditional dual-use technologies where export controls can restrict physical transfers, digital AI models can proliferate globally once released, making initial release decisions critically important for systems with significant capabilities.

Technical approaches to prevent misuse of open-weight models have often failed against determined actors. Security filters, output restrictions, and other measures implemented at the model level can be circumvented through techniques like prompt engineering, fine-tuning, or direct manipulation of model weights. Currently, security measures in open-weight models are easily circumvented and releases cannot be undone; open source models can be exploited by malicious actors.

Yet, open source models can be much more quickly incorporated into economic activities. DeepSeek's models have been rapidly incorporated into many popular applications in China, for instance. We can expect such open source models, and the applications that make use of them, to spread quickly around the world.

Control of the open source ecosystem by foreign actors creates a security vulnerability for the United States. Such models could include backdoors that are virtually undetectable, but triggered in contexts that are detrimental to U.S. security such as facilitating cyber security vulnerabilities.

The capability gap between open and closed models is surprisingly narrow, with recently released models demonstrating that open-weights AI can achieve capabilities similar to recently released proprietary systems—including potentially dangerous capabilities such as those relevant to chemical, biological, radiological, and nuclear threats.

Without a coherent framework, the U.S. risks allowing advanced systems and capabilities to proliferate without appropriate safeguards, enabling sophisticated threat actors—both non-state actors and foreign adversaries—to leverage American AI systems against American interests. The absence of clear guidelines about which capabilities should remain restricted and which can be openly shared creates uncertainty for developers while potentially allowing sensitive capabilities to proliferate unchecked. Indiscriminate release of unsecured models could provide adversaries with advanced technological capabilities they could not otherwise develop independently.

The United States should leverage its leading position in AI to shape the global open source ecosystem. It can promote open source with security guardrails, ensuring that default models used in the open source ecosystem do not undermine U.S. security interests. This means investing strategically in research on techniques to create open weight models that are more difficult to misuse, with particular attention to preventing the development of dangerous CBRN capabilities in open-weights models.

Open-source development remains extremely valuable and should be actively encouraged in numerous AI domains, including security techniques, evaluation benchmarks, model architectures, training methodologies, and specialized non-general purpose systems that present minimal misuse potential. These areas benefit from collaborative development while posing minimal security risks, allowing the innovation benefits of open approaches without the potential harms associated with unrestricted release of high-capability general-purpose models. Furthermore, many of the benefits commonly attributed to open-source AI—including external oversight, innovation acceleration, and democratic access—can be fully realized without the security risks of full release of model weights, such as through structured access programs and targeted API access ([Seger et al., 2023](#)). These approaches would preserve the benefits of transparency and accessibility for legitimate researchers and developers while maintaining appropriate controls on the most sensitive capabilities.

Recommendations:

- Partner with U.S. open source model providers to ensure that the open source ecosystem is not dominated by foreign actors.
- Invest in research to mitigate security-related misuse of open weight models.
- Create structured access frameworks for research and auditing of more capable models that preserve many benefits of openness (innovation, oversight, adaptation) while reducing proliferation risks.
- Restrict the release of model weights for highly capable systems that could enable chemical, biological, radiological, and nuclear (CBRN) threats or other catastrophic misuse. If such models are legal, such as white hat cyber tools, they should be made accessible only via secured APIs with appropriate safeguards and monitoring.
- Provide funding and resources for security research on open-source models to proactively identify and address vulnerabilities before they can be exploited, recognizing that market forces will not supply such research by default and that most adopting organizations lack the expertise and computing resources to evaluate these tools despite depending on them.
- Support the development of industry standards for evaluating when model components (e.g. weights) should be released publicly, based on specific capabilities and potential for misuse.

Conclusion

The Oxford Martin AI Governance Initiative believes that the recommendations outlined in this comment would help achieve the vision set forth in Executive Order 14179: removing barriers to American leadership in AI while promoting human flourishing, economic competitiveness, and national security. We appreciate the opportunity to contribute to this important discussion and stand ready to provide any additional information that might be useful in the development of the AI Action Plan.

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.